

## SECSR2043 OPERATING SYSTEMS

### [ 20 Marks]

Name : Poh Lok Yee  
 Student : A23CS0262  
 ID : \_\_\_\_\_  
 Section : Section 02

Marks

**Instruction:** Please answer all of the following questions. Whenever the 🖐 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: *ahmad.sh*).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.sh .; cp ../fruits/*.sh .;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]

**Print screen the script that you type;**

```
polok@secr2043: ~
GNU nano 7.2 pohlokyee.sh
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd drinks; cp ../cars/*.sh .; cp ../fruits/*.sh .;
cp ../*.jar .
```

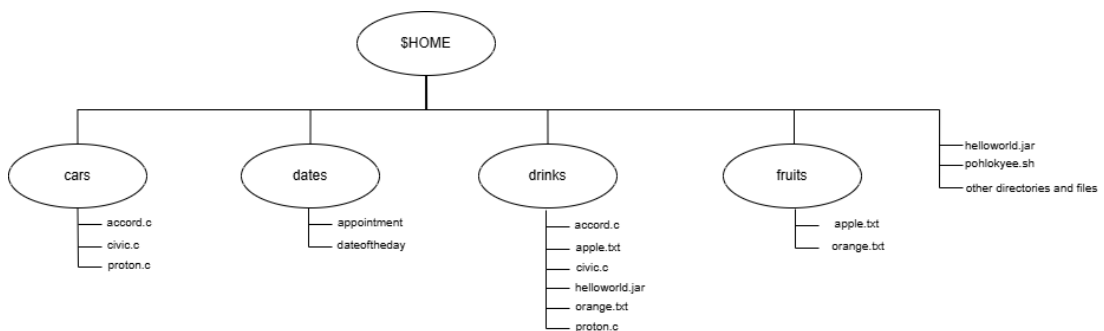
```

polok@secr2043:~$ nano pohlokyee.sh
polok@secr2043:~$ bash pohlokyee.sh
polok@secr2043:~$ tree
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   '-- proton.c
|-- dates
|   |-- appointment
|   '-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   '-- proton.c
|-- fruits
|   |-- apple.txt
|   '-- orange.txt
|-- helloworld.jar
|-- pohlokyee.sh
'-- snap
    '-- tree
        |-- 54
        |-- common
        |   '-- marker_publisher_changed_announcement_shown
        '-- current -> 54

10 directories, 16 files

```

Then draw the tree



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c

program files, and enter “c” as the input to the bash script.

[4 marks]

Print screen the bash script you type and run

```

polok@secr2043: ~
GNU nano 7.2 ifiles.sh
#!/bin/bash

#prompt user to enter file extension
read -p "Enter file extention without the dot:" ext

#finds and display all files wth given extention
echo "Files with ${ext} extension:"
find . -type f -name "*.${ext}"

#count number of files with the given extension
count=$(find . -type f -name "*.${ext}" | wc -l)
echo "Number of files with *.${ext} extension: $count"

polok@secr2043:~$ nano ifiles.sh
polok@secr2043:~$ bash ifiles.sh
Enter file extention without the dot:c
Files with .c extension:
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Number of files with *.c extension: 6

```



2. The following Figure 1 illustrates a tree structure of some directories and files.

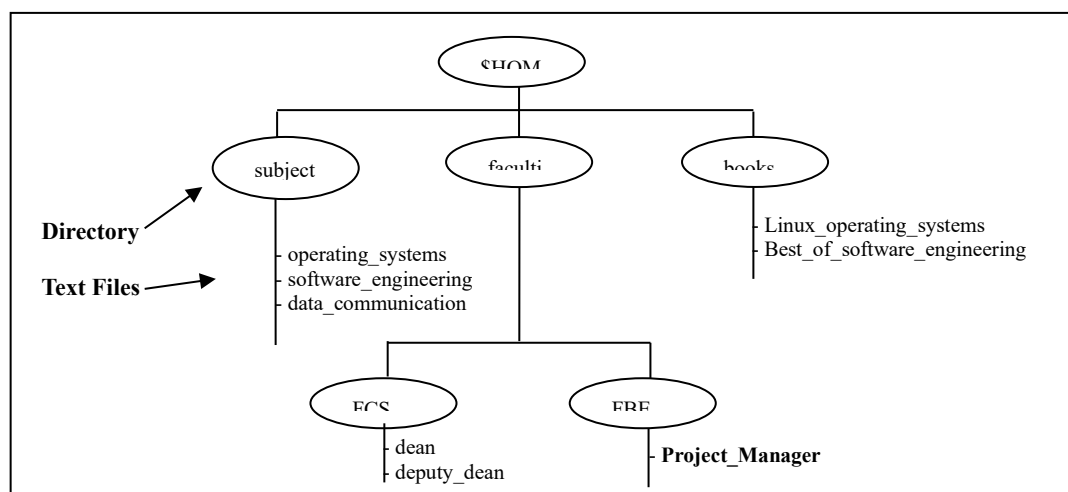


Figure 1

- a) Write a bash script (called myname2a.sh) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore

character. For example: text file Project\_Manager contains Project Manager).

[4 marks]

Print screen the bash script you type and run

```

polok@secr2043: ~
GNU nano 7.2                                pohlokyee2a.sh
#!/bin/bash

#create directories
mkdir -p $HOME/subjects $HOME/faculties/FCS $HOME/faculties/FBE $HOME/books

#create and write to text files in subject directory
echo "operating system" >$HOME/subjects/operating_system
echo "software engineering" >$HOME/subjects/software_engineering
echo "data communication" >$HOME/subjects/data_communication

#create and write to text files in the faculties/FCS directory
echo "dean" >$HOME/faculties/FCS/dean
echo "deputy dean" >$HOME/faculties/FCS/deputy_dean

#create and write to text files in the faculties/FBE directory
echo "Project Manager" >$HOME/faculties/FBE/Project_Manager

#create and write to text files in books directory
echo "Linux operating system" >$HOME/books/Linux_operating_system
echo "Best of software engineering" >$HOME/books/Best_of_software_engineering

polok@secr2043:~$ nano pohlokyee2a.sh
polok@secr2043:~$ bash pohlokyee2a.sh
polok@secr2043:~$ tree $HOME
/home/polok
|-- books
|   |-- Best_of_software_engineering
|   `-- Linux_operating_system
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- faculties
|   |-- FBE
|   |   `-- Project_Manager
|   `-- FCS
|       |-- dean
|       `-- deputy_dean
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
|-- ifiles.sh
|-- pohlokyee.sh
|-- pohlokyee2a.sh
|-- snap
|   `-- tree
|       |-- 54
|       |-- common
|       |   `-- marker_publisher_changed_announcement_shown
|       `-- current -> 54
`-- subjects
    |-- data_communication
    |-- operating_system
    `-- software_engineering
  
```



- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book.

[2 marks]

| Directory/File               | Access Control |
|------------------------------|----------------|
| books                        | drwxr-xr-x     |
| subjects                     | drwxr-xr-x     |
| Best_of_software_engineering | -rw-r--r--     |
| FCS                          | drwxr-xr-x     |
| project_manager              | -rw-r--r--     |

```

polok@secr2043:~$ ls -ld books
drwxr-xr-x 2 polok polok 4096 Jun 22 14:52 books
polok@secr2043:~$ ls -ld subjects
drwxr-xr-x 2 polok polok 4096 Jun 22 14:52 subjects
polok@secr2043:~$ ls -ld books/Best_of_software_engineering
-rw-r--r-- 1 polok polok 29 Jun 22 14:52 books/Best_of_software_engineering
polok@secr2043:~$ ls -ld faculties/FCS
drwxr-xr-x 2 polok polok 4096 Jun 22 14:52 faculties/FCS
polok@secr2043:~$ ls -ld faculties/FBE/Project_Manager
-rw-r--r-- 1 polok polok 16 Jun 22 14:52 faculties/FBE/Project_Manager

```

- c) Write another bash script (called myname2c.sh) that will change the access control of the directories and files based on the following information:

[4 marks]

| Directory/File               | Users |   |   |       |   |   |        |   |   |
|------------------------------|-------|---|---|-------|---|---|--------|---|---|
|                              | Owner |   |   | Group |   |   | Public |   |   |
| subjects                     | ✓     | ✓ | ✓ | ✓     | x | x | ✓      | x | x |
| Best_of_software_engineering | ✓     | x | ✓ | x     | ✓ | x | x      | x | x |
| FCS                          | ✓     | ✓ | x | x     | x | x | ✓      | ✓ | ✓ |
| project_manager              | x     | x | x | x     | ✓ | ✓ | x      | x | ✓ |

Print screen the bash script you type and run

```

polok@secr2043: ~
GNU nano 7.2 pohlkyee2c.sh *
#!/bin/bash

#set permission for subject directory
chmod 744 $HOME/subjects

#set permission for Best_of_software_engineering file
chmod 520 $HOME/books/Best_of_software_engineering

#set permission for FCS directory
chmod 607 $HOME/faculties/FCS

#set permission for project_manager file
chmod 031 $HOME/faculties/FBE/Project_Manager

polok@secr2043:~$ nano pohlkyee2c.sh
polok@secr2043:~$ bash pohlkyee2c.sh
polok@secr2043:~$ ls -ld ~/subjects
drwxr--r-- 2 polok polok 4096 Jun 22 14:52 /home/polok/subjects
polok@secr2043:~$ ls -ld ~/faculties/FCS
drw----rwx 2 polok polok 4096 Jun 22 14:52 /home/polok/faculties/FCS
polok@secr2043:~$ ls -ld ~/books/Best_of_software_engineering
-r-x-w---- 1 polok polok 29 Jun 22 14:52 /home/polok/books/Best_of_software_engineering
polok@secr2043:~$ ls -ld ~/faculties/FBE/Project_Manager
-----wx-x 1 polok polok 16 Jun 22 14:52 /home/polok/faculties/FBE/Project_Manager

```

- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]



| Directory/File               | Access Control |
|------------------------------|----------------|
| subjects                     | drwxr--r--     |
| Best_of_software_engineering | drw----rwx     |
| FCS                          | -r-x-w----     |
| project_manager              | -----wx--x     |

End of Lab 3

\*\*\* All the Best for Final Exam \*\*\*